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4201 L
Lab 10

Lab 9 part 1: Single Supply Op Amps for Intrumentation

Introduction

The objective of this laboratory is to understand some of the differences, limitations and usage for op amps in Single Power supply circuits (+Vcc and Com/Gnd typically) compared to traditional dual supply (+Vcc –Vcc Com/Gnd implicit) circuits. We have already worked with amplifier circuits (inverting amplifiers, non-inverting amplifiers, voltage followers, summers, integrators, and differentiators). Most of these previous labs used dual supply op amps (which operate with both positive and negative supply voltages and follow familiar and well develoiped equations). However, many modern electronic systems, (such as the ADC circuits we studied, thermistor circuits, microcontrollers, etc) operate using only a single power supply and do not have access to negative supply voltages.When an op amp is powered by a single supply, its behavior changes significantly. The goal of this lab is to understand how a single supply op amp works

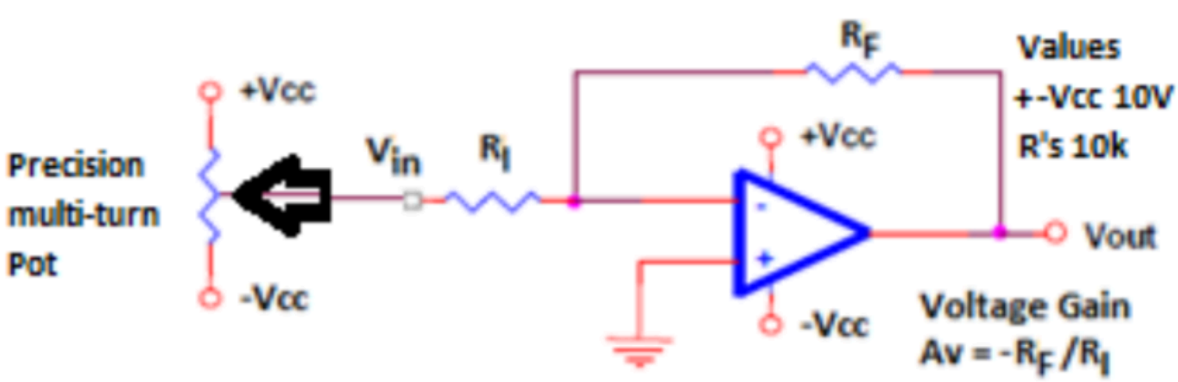
For example, consider a simple inverting amplifier with a gain of −1. If the supply voltage is 10 V and the input voltage is +2 V, the expected output from the gain equation would be −2 V. However, a single-supply op amp cannot produce a negative output voltage because its output is limited to the range between ground (0 V) and the positive supply voltage. As a result the output becomes stuck at 0 V, and the circuit no longer behaves as expected. If the input voltage is negative, the output may fall within the allowable range, but the behavior becomes difficult to predict because tying the non-inverting input directly to ground affects the internal operation of the op amp. This makes single-supply op amp circuits more complex and less reliable if traditional designs are used without modification.

To address this issue the lab focuses on understanding the operational voltage limits of single-supply op amps and introduces a simple and practical solution. Instead of referencing the op amp to ground, a new reference voltage is created at the non-inverting input. This reference voltage is typically set to half of the supply voltage (Vcc/2), creating what is known as a “virtual ground.” With this virtual ground the op amp output can swing both above and below the reference point while still remaining within the allowable supply range. This reference voltage can be created using two equal-value resistors connected between Vcc and ground, with their midpoint connected to the non-inverting input. Although more advanced solutions such as precision rail-splitter devices exist, the resistor divider method is sufficient for this lab and allows single-supply op amp circuits to behave in a predictable and useful way.

Part 1

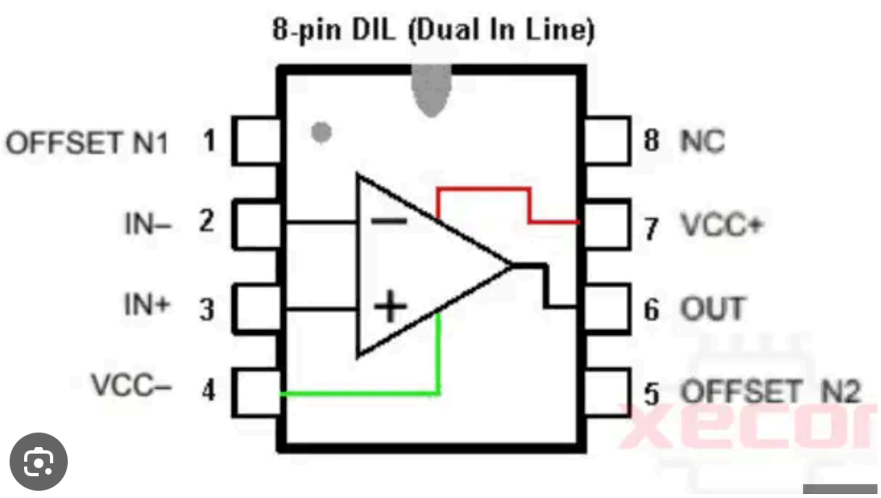
1) Build a conventional dual-supply inverting amplifier with a gain of -1, i.e. RF = RI as shown in image 1. Use your potentiometer from previous labs (precision 10 turn) with each end connected to +Vcc and –Vcc so Vin can be adjusted between +Vcc and –Vcc. Measure and plot Vout vs. Vin as Vin changes 1V at a time. Check and confirm gain in linear, non-saturated, region of data. Determine the appropriate range of input voltages, i.e. those the result in a proportional output response.

Components:
- UA741CP operational amplifier
- supply voltages +10V and -10V
- Rf and R1 of 10kΩ

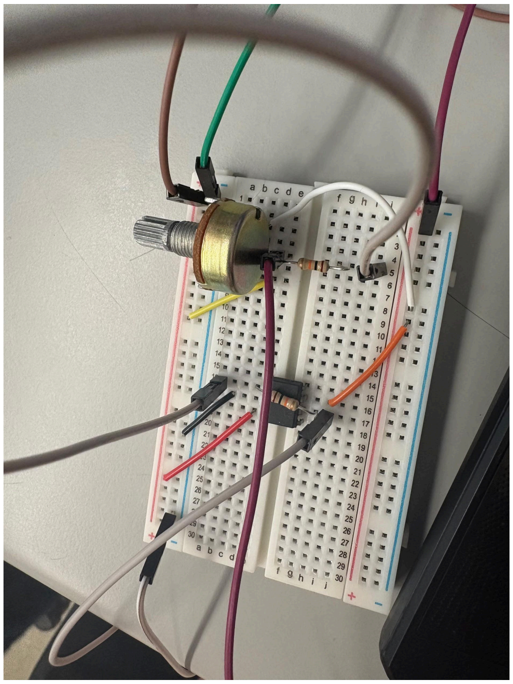


IMG 1. *Circuit schematic*

IMG 2. *Supply voltages +10V and -10V*



IMG 3. *UA741CP operational amplifier pinout, found at* https://www.xecor.com/blog/ua741cp-pinout-circuit-equivalent-datasheet-ua741cp-vs-lm741

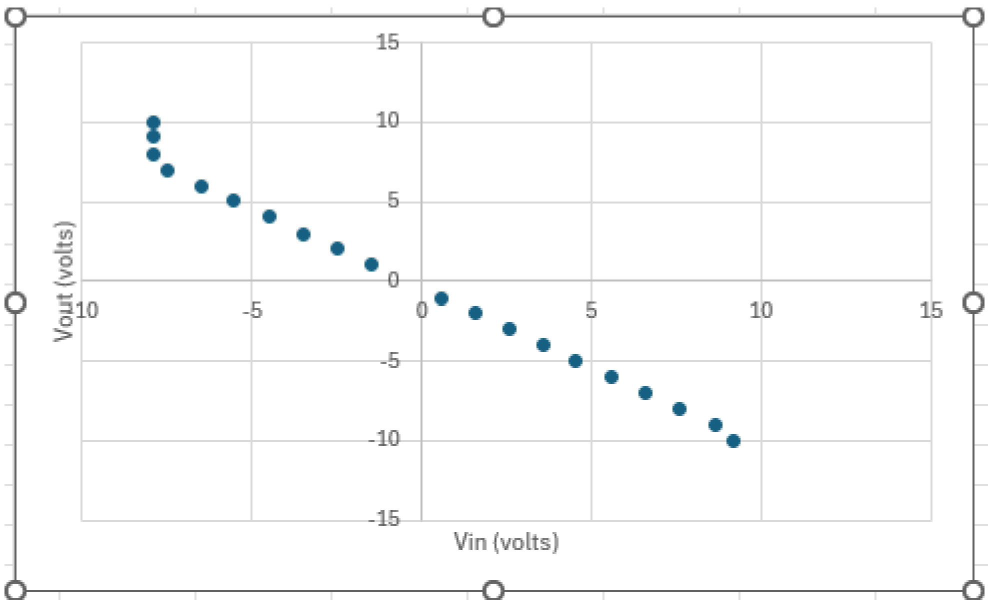


IMG 4. *Physical circuit for part 1*

Vout (v)	Vin (v)
9.18	-9.97
8.66	-9.07
7.61	-8.01
6.59	-7.02
5.57	-6.00
4.55	-5.00
3.584	-4.005
2.591	-3.027
1.572	-2.001
0.571	-1.03
-1.48	1.00
-2.48	2.01
-3.48	3.00

-4.5	4.02
-5.55	5.05
-6.51	6.01
-7.50	7.00
-7.91	8.01
-7.91	9.08
-7.91	9.97

Table 1. *Measured vout for increments of 1v of Vin*

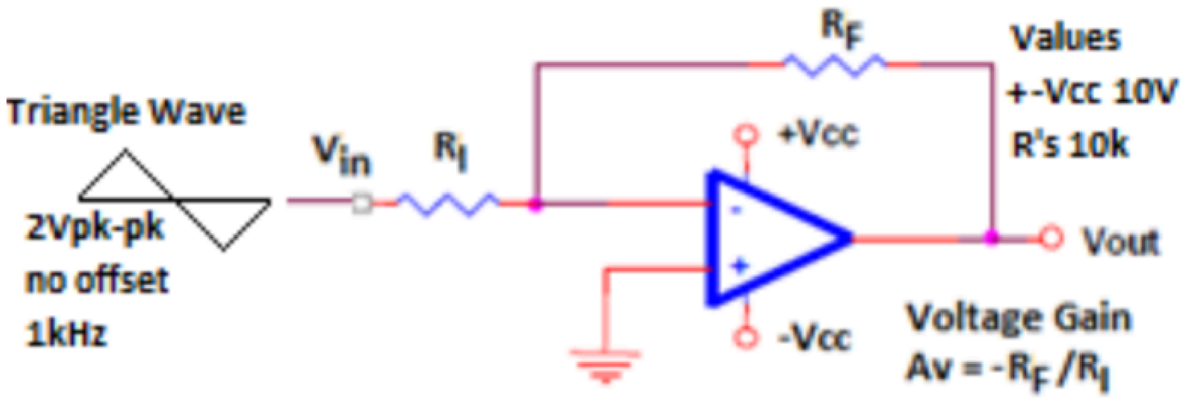


IMG 5. *Data is accessible here: [Single supply op amps data.xlsx]*

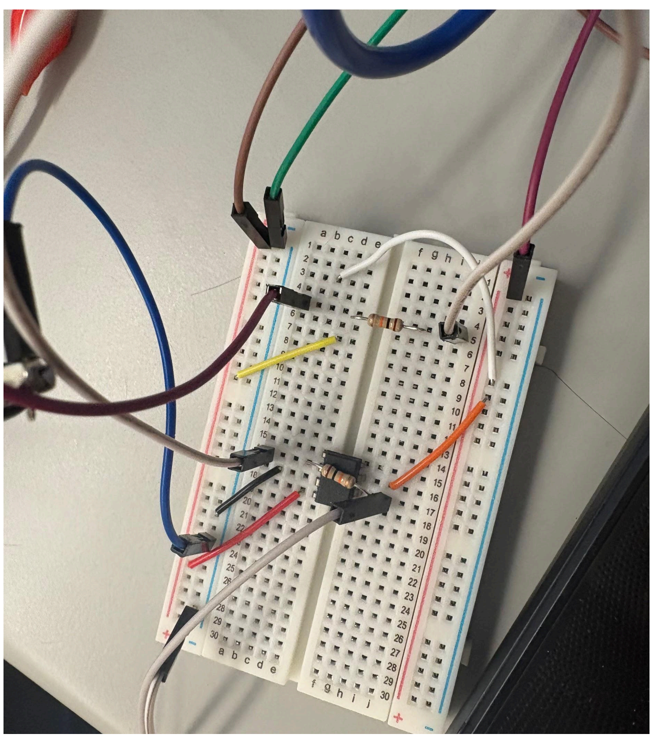
Appropriate range of input voltages resulting in a proportional output response:
approximately -9.97 volts to 8.01v

Part 2

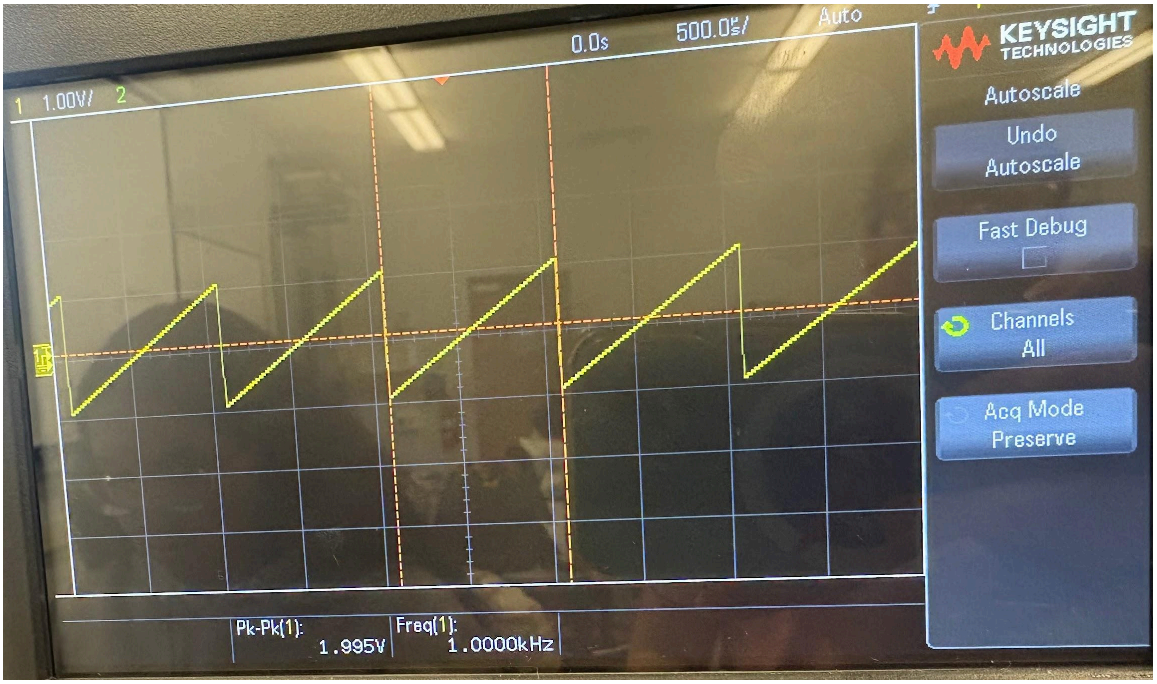
2) A) Remove the precision pot as the input voltage and insert a 2Volt peak to peak triangle wave as an input signal and record/capture Vout and Vin, a sine wave could also be used but triangle waves often show linearity issues easier. B) Increase amplitude of triangle wave until noticeable distortion appears in the output signal, record output signal and (+) reference input with noticeable distortion. If the generator cannot output enough signal see the instructor or change the output to high Z mode via the menus. Explain what is happening with (+) reference signal when the output signal becomes distorted. Does the output response match appropriate range of input voltages from part 1?



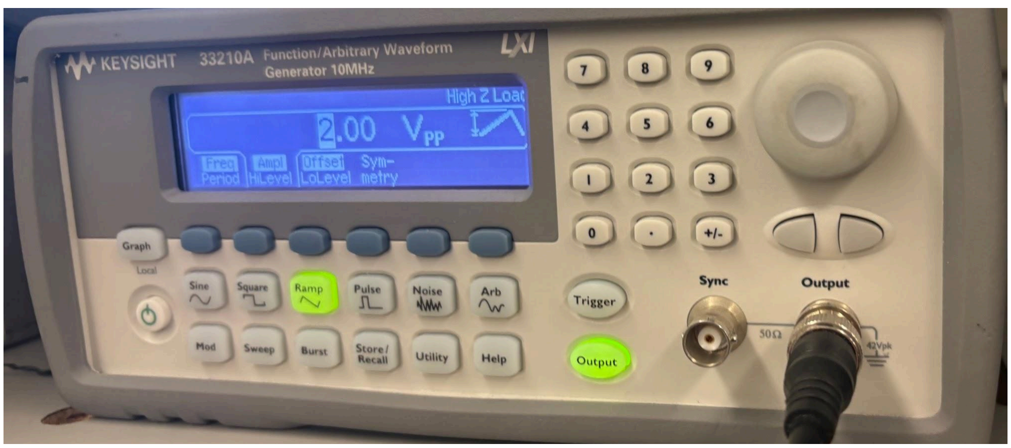
IMG 6. *Circuit schematic*



IMG 7. *Physical circuit*



IMG 8. *2 volt peak to peak triangle wave*

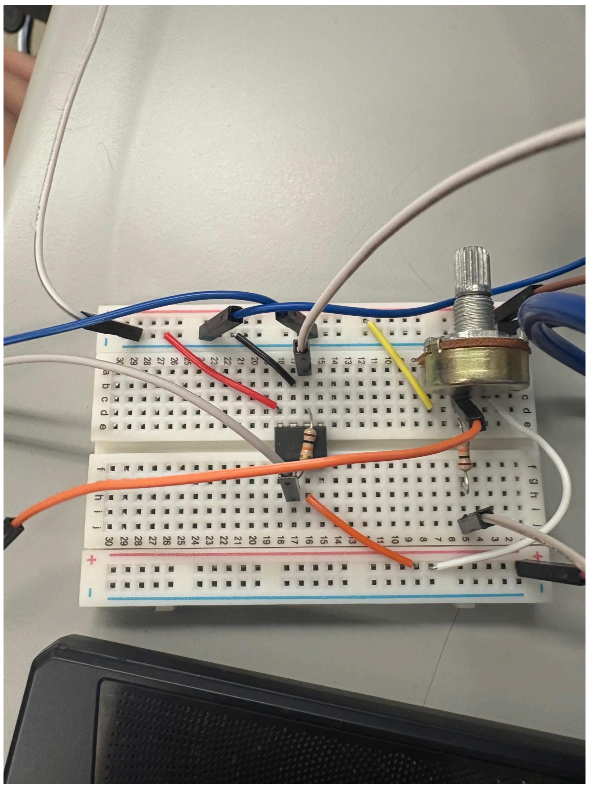


IMG 9. *2 volt peak to peak triangle wave*

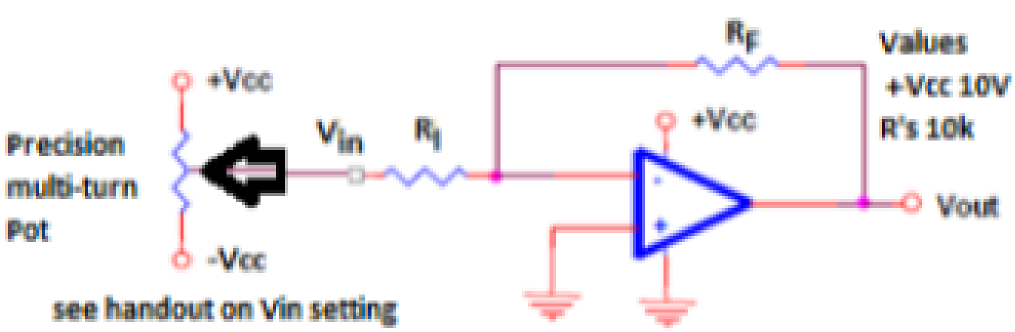
The output response mostly matches the expected range of input values from part 1. This shows that single-supply op amps can work for the application and are adequate but they are not ideal (single supply op amps particularly so). Their ± behavior only operates correctly around the reference voltage (not around 0 V) which introduces limitations and imperfections in the response.

Part 3

Change circuit to single supply i.e. –Vcc becomes Gnd, samevalues for circuit, set your potentiometer for a negative voltageof -2Volts to start data collection, monitor and record Vout as Vinis changed negatively and positively from -2Volts, when Vout nolonger changes as Vin changes stop data collection. As in part 1 determine the appropriate range voltages



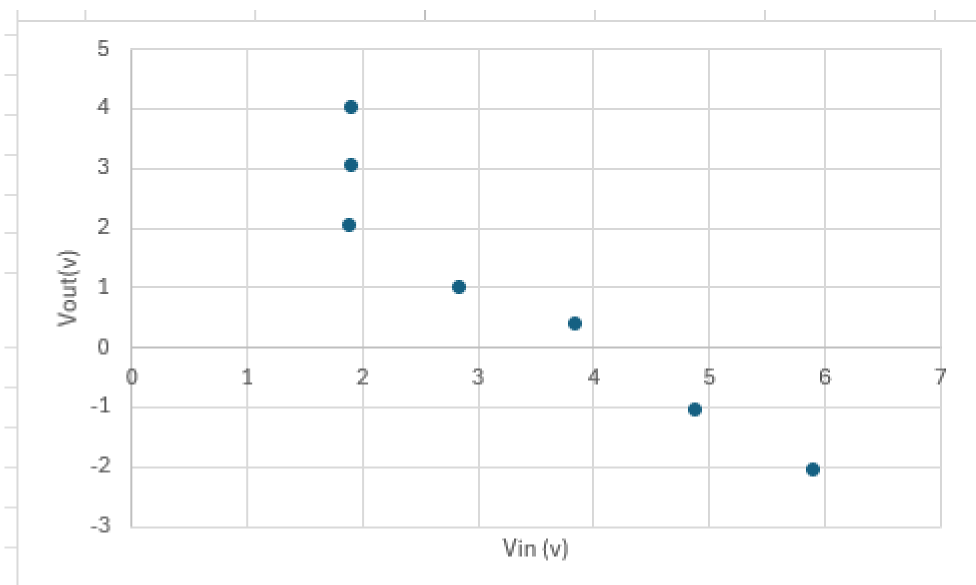
IMG 10. *Physical circuit*



IMG 11. *scbematic*

Vout (volts)	Vin (volts)
5.89	-2.045
4.87	-1.038
3.836	0.4
2.830	1.001
1.89	2.039
1.9	3.069
1.9	4.03

Table 2.

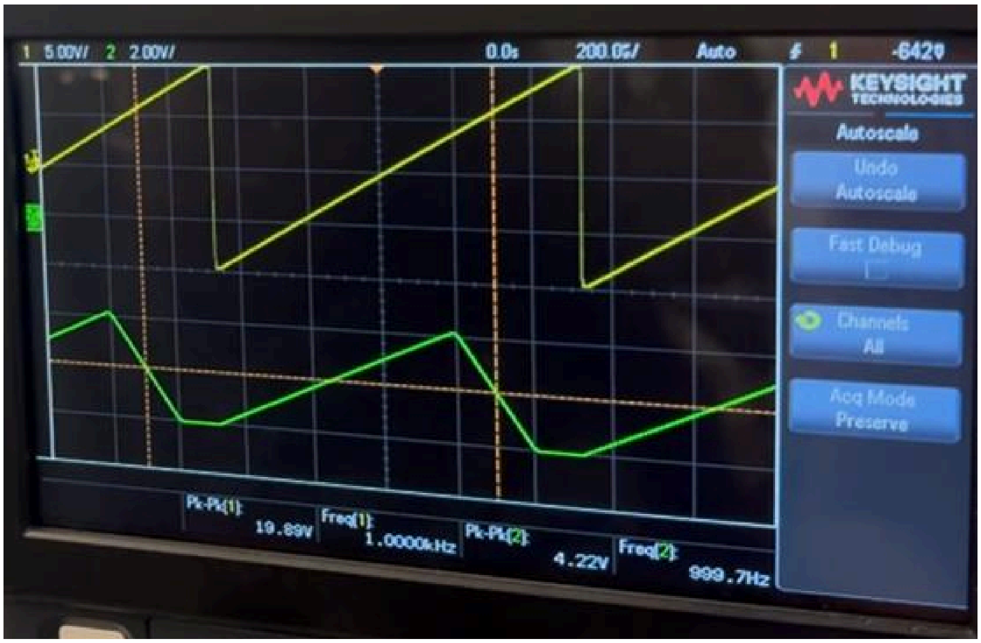


IMG 12 . *Data is accessible here: [Single supply op amps data.xlsx]*

Appropriate range of input voltages resulting in a proportional output response: *approximately -2.04 volts to 4.03*

Part 4

As with circuit 2) use a triangle wave signal with single-supply circuit 3) to view Vin and Vout, you willneed to adjust the amplitude and offset of the signal to obtain a signal that looks good with no distortion.B) Adjust offset and amplitude to get maximum signal out without distortion, record result. Does theoutput response match appropriate range of input voltages from part 3?

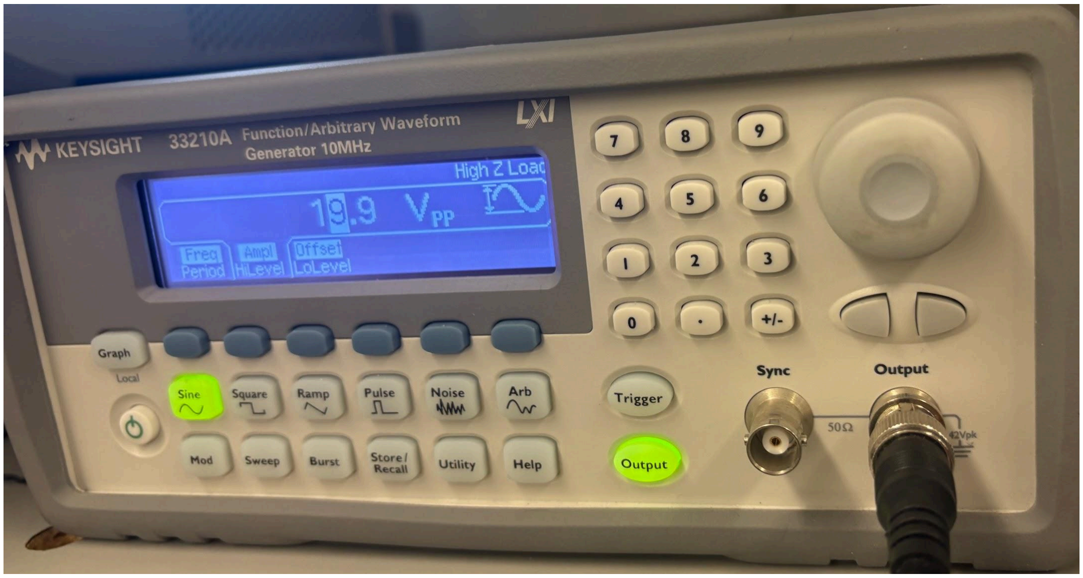


IMG 14 . *Distortion is visible*

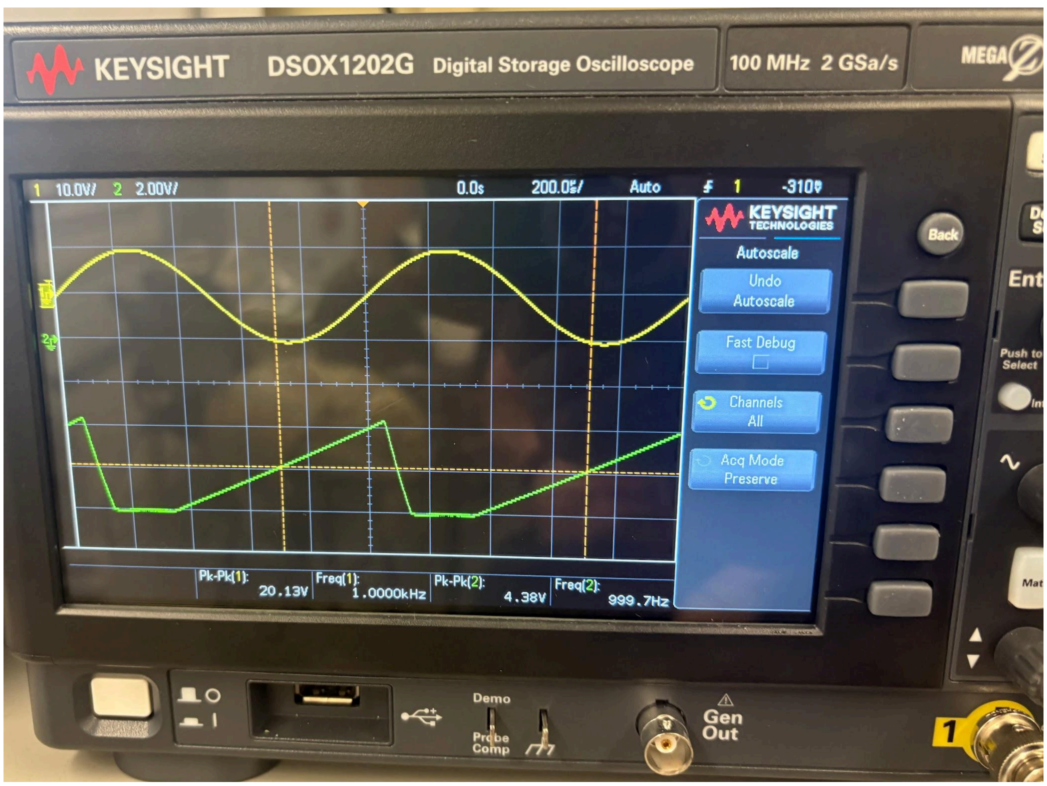
Video of triangle wave distortion: https://youtu.be/Dbl0yc4dKkM

The input signal (Channel 1, yellow) and the output signal (Channel 2, green) generally match in shape, but the output highlights that op amps are not as ideal as theory suggests which is an important consideration in instrumentation. The performance is adequate, though not perfect. The output is clearly inverted, which is expected because the circuit uses an inverting op-amp configuration. We can also see that the output response approximately matches the appropriate range of input voltages from part 3.

Distortion for sine wave:


KEYSIGHT
33210A Function/Arbitrary Waveform Generator 10MHz
LXI
High Z Load
19.9 Vpp
Freq
Period
Ampl
HiLevel
Offset
LoLevel
Graph
Local
Sine
Square
Ramp
Pulse
Noise
Arb
Mod
Sweep
Burst
Store/Recall
Utility
Help
Trigger
Output
Sync
Output
50Ω

IMG 15.


KEYSIGHT
DSOX1202G Digital Storage Oscilloscope
100 MHz 2 GSa/s
1 10.0V/ 2 2.00V/
0.0s
200.0$/
Auto
1 -310
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels All
Acq Mode Preserve
Back
Pk-Pk(1): 20.13V
Freq(1): 1.0000kHz
Pk-Pk(2): 4.38V
Freq(2): 999.7Hz
Demo
Probe Comp
Gen Out

IMG 16.

5.

In this lab we tested three different op-amp configurations and compared the measured Vout versus Vin behavior to the expected transfer functions. In each case, the “transfer function” is the slope of the straight-line region of the Vout vs Vin plot (i.e. the gain) together with any offset.

1.Dual-Supply Inverting Amplifier
For the conventional inverting amplifier with dual supplies (+Vcc and –Vcc), the theoretical transfer function is the standard inverting-amp equation:$v_{out} =- \frac{Rf}{RI} V_{in}$. However, since $R_F = R_I$ the ideal transfer function is actually just $V_{out} = - V_{in}$ or $- 1 = \frac{V_{out}}{V_{in}}$.

Vout (v) (y axis)	Vin (v) (x axis)
9.18	-9.97
8.66	-9.07
7.61	-8.01
6.59	-7.02
5.57	-6.00
4.55	-5.00
3.584	-4.005
2.591	-3.027
1.572	-2.001
0.571	-1.03
-1.48	1.00
-2.48	2.01
-3.48	3.00
-4.5	4.02
-5.55	5.05
-6.51	6.01
-7.50	7.00
-7.91	8.01

Table 3. *Information from table 1, data from part 1*

$\frac{\Delta Y}{\Delta X} = \frac{9.18-(-7.91)}{-9.97-8.01} = \frac{17.09}{-17.98} =- 0.95$

-0.95 is pretty close to -1. This confirms that the appropriate range was identified and confirms the transfer function validity and aligns with the scope readings.

2. Single-Supply Using the Same Op Amp (0 to +Vcc, no rail-splitter)

Next we switched to single-supply operation by replacing –Vcc with ground while keeping the same inverting configuration. Mathematically the ideal equation is still $v_{out} =- \frac{Rf}{RI} V_{in}$. However, in single-supply operation the op amp output is limited to approximately 0 V to +VCC. Any portion of the ideal transfer function that would require the output to go below 0 V or above +VCC cannot be realized, causing the practical transfer function to become piecewise. For a small range of input voltages where the ideal output remains within the allowable range, the response appears approximately linear with a slope near −1. Outside of this region, the output saturates at either the lower rail (near 0 V) or the upper rail (near +VCC), and further changes in the input no longer affect the output. Experimentally, sweeping the input voltage in both the negative and positive directions showed that the output quickly reached one of the supply rails and then remained nearly constant. This demonstrates that tying the non-inverting input directly to ground in a single-supply system forces the amplifier to operate at the edge of its allowable output range, resulting in a very limited and highly asymmetric linear region. As a result, waveform plots show significant clipping rather than a clean, symmetric response, and the simple relationship Vout=−Vin is only valid over a very narrow input range due to the inability of the op amp to produce negative output voltages on a 0 to +VCC supply.

3. Single-Supply with Rail-Splitter (Virtual Ground at Vref = Vcc/2, TLC27L4)

Finally, we used a proper single-supply op amp (TLC27L4) and created a “virtual ground” at:
$v_{red} = \frac{Vcc}{2}$. with a resistor divider (and bypass capacitor). The non-inverting input (+) is connected to this Vref instead of to 0 V. In the inverting configuration, the op amp forces the inverting input (–) to be approximately equal to the non-inverting input, so the inverting input node also sits near Vref.

Writing KCL at the inverting input node gives the exact transfer function:

$$\frac{V_{out} - V_{ref}}{R1} + \frac{V_{out} - V_{ref}}{RF} = 0$$

Solving for Vout gives us

$$V_{out} - V_{ref} = - \frac{Rf}{R1} (V_{out} - V_{in})$$

Or

$$V_{out} = V_{ref} - \frac{Rf}{R1} (V_{in} - V_{ref})$$

And when Rf=R1, we get

$$V_{out} = V_{ref} - (V_{in} - V_{ref}) = 2V_{ref} - V_{in}$$

The key idea is that the small-signal gain (the slope of the transfer function) is still equal to −RF/RI, which is approximately −1 in this case, but the entire response is now centered around a reference voltage, Vref, instead of 0 V. In the measurements, the Vout versus Vin plot in the linear operating region again formed a straight line with a slope close to −1; however, this line was centered at Vout = Vref when Vin = Vref. In the time domain, the output voltage waveforms for a triangular input appeared roughly symmetric around Vref, which was approximately VCC/2, rather than around ground. By creating this “virtual ground” at mid-supply, the op amp has voltage headroom both above and below Vref, allowing the output to swing without immediately hitting 0 V or VCC. As a result, the amplifier operates over a much larger and more symmetric linear region, similar to the original dual-supply case, while using only a single positive supply.

Conclusion

In this lab we examined how operational amplifiers behave under both dual-supply and single-supply conditions by comparing their transfer functions. The dual-supply inverting amplifier produced a nearly ideal linear relationship with a gain close to –1 and a symmetrical output centered around 0 V. This confirmed the expected transfer function

$v_{out} =- \frac{Rf}{RI}V_{in}$.When the same amplifier was powered from a single supply and referenced to ground, the output quickly reached the 0 V rail.. Although the theoretical transfer function remains the same, the practical linear region became extremely limitedbecause the op amp could not produce negative output voltages. This resulted in significant distortion in the measured waveforms and Vout–Vin plots.

Finally, by creating a virtual ground at $v_{red} = \frac{Vcc}{2}$. and using a rail-to-rail single-supply op amp we restored a large and symmetric linear region. The measured transfer function again showed a slope close to –1 but now centered around Vref instead of 0 V. This allowed the amplifier to swing both above and below the midpoint without immediately saturating against the supply rails. Overall, the results demonstrate that proper reference voltage selection is essential in single-supply op-amp design. Establishing a virtual ground allows a single-supply amplifier to closely replicate the behavior of a dual-supply configuration, providing both symmetry and adequate headroom for linear operation. This lab reinforced that while the ideal transfer function remains mathematically the same, the real-world output range and distortion depend heavily on supply configuration and reference biasing.

Proving transfer functions:
https://docs.google.com/document/d/19BJ1pcpWYcYVTunoJ9pc3xC8CnAgnqG6ivv5aELVo6w/edit?usp=sharing